FOR AN IRREDUCIBLE SUBSET AND A FINITE UNION OF LOCALLY CLOSED SUBSETS THE THREE DENSITY CONDITIONS ARE EQUIVALENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of irreducibility and of a constructible (equivalently finite union of locally closed) subset, and the notion of a generic point. Set up in the author's sections on irreducible components and on constructible sets, before the counted statement. Fixes the three conditions whose equivalence the statement asserts.

Lemma 0.1. *Let X be a topological space. Suppose that $Z \subset X$ is irreducible. Let $E \subset X$ be a finite union of locally closed subsets (e.g. E is constructible). The following are equivalent*

(1) *The intersection $E \cap Z$ contains an open dense subset of Z.*
(2) *The intersection $E \cap Z$ is dense in Z.*

If Z has a generic point ξ, then this is also equivalent to

(3) *We have $\xi \in E$.*

Proof. The implication (1) $\Rightarrow$ (2) is clear. Assume (2). Note that $E \cap Z$ is a finite union of locally closed subsets Z_i of Z. Since Z is irreducible, one of the Z_i must be dense in Z. Then this Z_i is dense open in Z as it is open in its closure. Hence (1) holds.

Suppose that $\xi \in Z$ is a generic point. If the equivalent conditions (1) and (2) hold, then $\xi \in E$. Conversely, if $\xi \in E$ then $\xi \in E \cap Z$ and hence $E \cap Z$ is dense in Z. $\square$

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